Paper Name

Data Structure through OOP (A5-R5.1)

Chapter -1: Object Oriented Concepts

Object Oriented Programming- a new paradigm, Abstraction, forms of Abstraction, OOP concepts- Classes, Objects, Polymorphism, Data Encapsulation, Data Hiding,Inheritance,

Chapter -2: Basics of C++, Classes and Objects

Features of C++, Tokens, keywords, Data types, Operators, Manipulators, Console input, output, Control statements (conditional and loops), Functions, Classes, Instantiation, Destructor, constructor, Polymorphism - Operator Overloading, Function Overloading, Inheritance-Single, Multiple, Multilevel, Pointers

Chapter -3: Analysis of Algorithm

Introduction to algorithm design and Data structures, Comparison of Algorithms, Complexity in terms of space and time, Calculation of O- notation. Abstract Data type and its implementation with a Rational number example

Chapter -4: Searching and Sorting

Searching- Linear and Binary Search, Sorting- Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, Merge Sort, Comparison of various searching and sorting techniques in terms of time complexity

Chapter -5: Elementary Data Structures: Arrays, Linked Lists

Representation of arrays-single and multidimensional, Address calculation using row major ordering, Various operations on arrays, Linked Lists-Singly Linked List, Double linked List, Circular Linked List- traversing, deleting, inserting, searching, counting, reversing, printing of nodes.

Chapter -6: Stacks and Queues

Stack ADT, Implementation of stack using array and linked list, Application of Stack- Evaluation of postfix/prefix expression, Queue ADT, Implementation of queue using Array and Linked List

Chapter -7: Trees

Definition and notations, Binary Search Trees Implementation. Traversals using stacks and recursion - In-order, post-order, pre-order techniques, Threaded binary tree, B-trees with implementation of 2-3 trees.

Chapter -8: Graphs

Definition and notations, Components of Graphs, Types of Graphs, Graph Implementation using Adjacency Matrix and Adjacency List algorithms and programs, Graph Traversal Methods: Depth First Search and Breadth First Search.

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